1.Title:

Personalized TED Talks Recommendation System Using Machine Learning.

2.Project Statement:

With the ever-growing library of TED Talks spanning diverse topics and speakers, users often struggle to discover talks that match their interests. To address this challenge, a recommendation system can help users efficiently explore talks tailored to their tastes, boosting engagement and learning. This project aims to develop a machine learning–based recommender system that suggests relevant TED Talks by analyzing content such as talk transcripts and tags, as well as user preferences and behavior patterns. By combining content-based and collaborative filtering techniques, the system will deliver personalized recommendations through an intuitive, user-friendly interface designed to enhance the discovery experience and encourage deeper exploration of ideas.

Outcomes:

The goal is to build a recommendation engine that:

- Understands and analyzes the content of TED Talks.
- Learns from user interactions and preferences.
- Generates personalized recommendations.
- Can be deployed in an accessible web-based app.

Modules to be implemented

- Data Collection
- Data Exploration (EDA) and Data Preprocessing
- Feature Engineering and Vectorization
- Develop Recommendation Engine
- Evaluation Metrics
- Deployment via Streamlit App
- Presentation and Documentation

3. Week-wise Module Implementation and High-Level Requirements with Expected Outputs

Milestone 1: Week 1-2

Module 1: Data Collection

- Collect a high-quality dataset of TED Talks including metadata such as titles, tags, speaker names, transcripts, and user ratings or watch history if available.
- Ensure diversity of topics and sufficient volume for effective training.
- Example sources: TED.com scraping, Kaggle TED Talks datasets, public transcript archives.

Module 2: Data Exploration and Data Preprocessing

• Inspect dataset for completeness, missing values, and inconsistencies.
• Clean text data (e.g., remove stopwords, punctuation, lowercase conversion).
• Handle missing transcripts or tags.
• Analyze distribution of topics, talk lengths, and other metadata.
• Encode categorical features (e.g., speaker, tags) as needed.
• Prepare user–item interaction matrix (for collaborative filtering if user data is available).
• Create initial content similarity metrics (e.g., text length, word counts).

Expected Output:

• Cleaned, preprocessed dataset ready for feature extraction and modeling.
• Visualizations of data distribution (talk lengths, tag frequencies).
• Summary statistics capturing data quality.

Milestone 2: Week 3-4

Module 3: Feature Engineering and Vectorization

• Convert text data (transcripts, tags) into numerical representations.
• Apply vectorization techniques such as TF-IDF or embedding models (Word2Vec, BERT).
• For collaborative filtering, construct and sparsely represent user–item interaction matrices.
• Document the choice of vectorization methods and parameter settings.
• Analyze and visualize feature importance or similarity scores for interpretability.

Expected Output:

• Saved TF-IDF / embedding models.
• Vectorized representations of talk transcripts and tags.
• User–item interaction matrix (if applicable).
• Similarity matrices for content-based filtering.

Module 4: Develop Recommendation Engine

- Choose and implement appropriate recommendation strategies:
- Content-based filtering (similarity using text features).
- Collaborative filtering (matrix factorization using user data).
- Hybrid approaches combining both signals.
  - Split data into training and testing sets for offline evaluation.
  - Train models and save them for deployment.
  - Document model training process, including hyperparameter choices and rationale.

Expected Output:

• Trained recommendation models (content-based, collaborative, or hybrid).
• Saved model files for reuse in deployment.
• Code notebooks/scripts demonstrating training workflow.

Milestone 3: Week 5-6

Module 5: Evaluation Metrics

• Evaluate recommendation quality using metrics such as:

- Precision
- Recall
- NDCG (Normalized Discounted Cumulative Gain)
- Mean Average Precision
  - Compare different recommendation approaches.
  - Analyze trade-offs between content-based and collaborative filtering.
  - Visualize evaluation results.
  - Select final model(s) based on balanced performance and interpretability.

Expected Output:

• Evaluation reports and plots.
• Comparison tables summarizing model performance.
• Documented justification for final model selection.

Milestone 4: Week 7-8

Module 6: Deployment via Streamlit App

• Design and develop an interactive web-based interface using Streamlit.
• Integrate trained recommendation models.
• Provide user input options (e.g., enter a previously liked TED Talk, select tags, user profile).
• Display top-N recommended TED Talks with details (title, speaker, tags, description, link).
• Optionally include content explanations (e.g., "because it's similar to X" or feature insights).
• Ensure responsiveness and user-friendliness.
• Document the full deployment process.

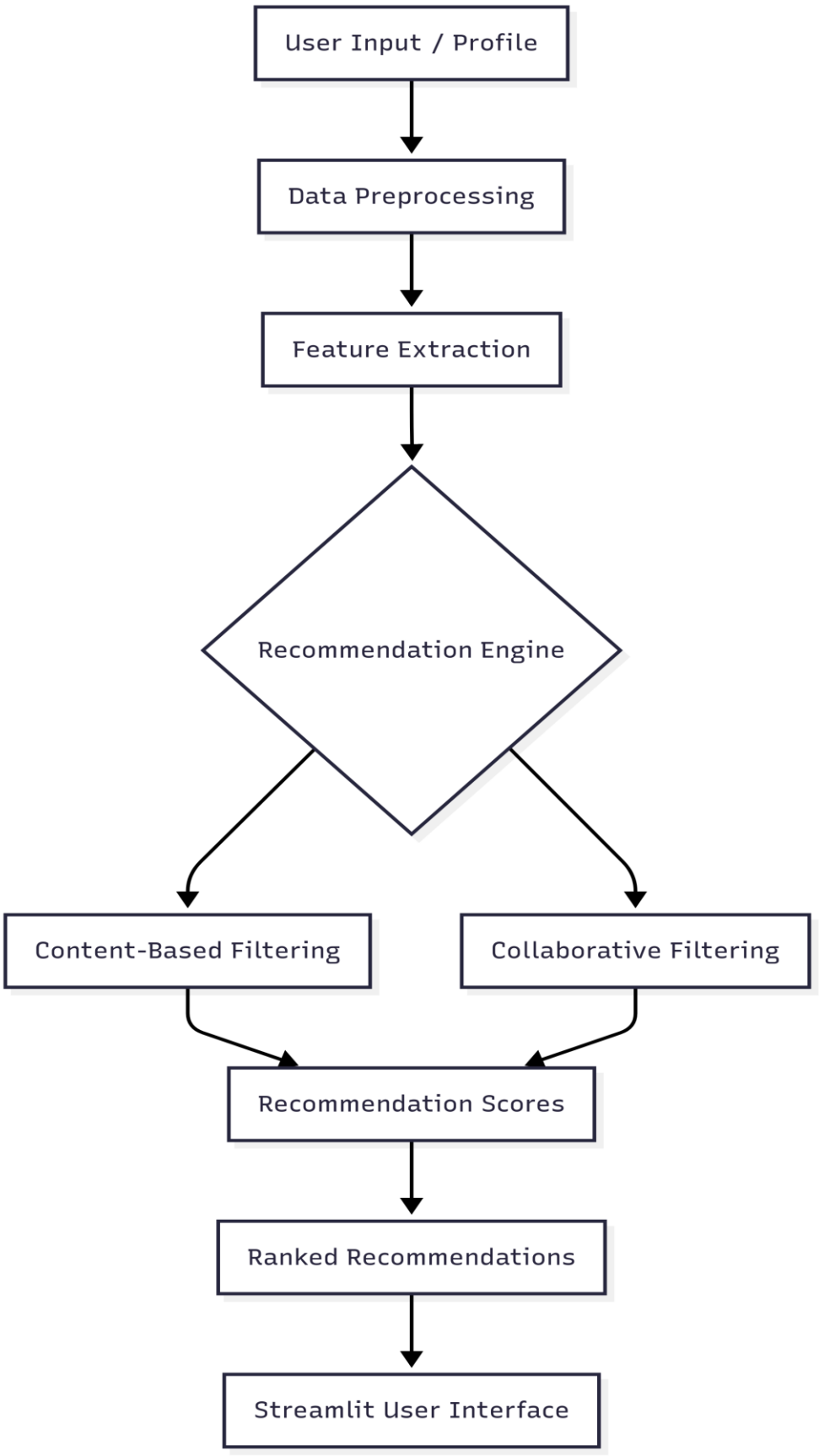
Expected Output:

• Fully functional Streamlit app.
• Screenshots of app interface with example recommendations.
• Documentation for deploying and running the app locally or on a server.

Module 7: Presentation and Documentation

• Prepare comprehensive project documentation covering all modules, methods, and design decisions.
• Include visualizations (EDA, feature importance, evaluation metrics).
• Capture screenshots of the deployed app interface.
• Create presentation slides summarizing key outcomes, methods, and visuals.
• Ensure clarity, completeness, and professional quality.

TED Talks Recommendation System Workflow



Evaluation Criteria:

Milestone 1 Evaluation (Week 2):

- Approval of collected TED Talks dataset, ensuring richness and diversity.
- Approval of exploratory analysis (e.g., talk length distributions, topic tags).
- Approval of preprocessing pipeline for cleaning transcripts and metadata.

Milestone 2 Evaluation (Week 3-5):

- Approval of vectorization techniques (TF-IDF, embeddings) with documented rationale.
- Approval of feature engineering choices and resulting representations.
- Approval of collaborative filtering setup and user–item matrix design.
- Approval of saved vectorizer and embedding models.

Milestone 3 Evaluation (Week 6-7):

- Approval of recommendation engine design and model selection (content-based, collaborative, or hybrid).
- Approval of evaluation metrics and analysis (Precision@K, Recall@K, NDCG).
- Approval of comparative analysis and final model selection rationale.

Milestone 4 Evaluation (Week 7–8):

- Approval of fully functional Streamlit web app delivering personalized TED Talk recommendations.
- Approval of app design, input options, and clarity of recommendation outputs.
- Approval of thorough deployment documentation.
- Approval of final project documentation and presentation slides.
- Approval of final code submission to GitHub or designated repository, ensuring reproducibility and clarity.